Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Examine the concept of a relational model and analyze its application in creating a database for managing student records. Differentiate between entity sets and relationships within this context.	**2**	(CO2, L4)
2)	Examine and analyze the different layers involved in software development, explaining how each layer contributes to the overall functionality of an ethical software product.	**2**	(CO1, L4)
3)	Examine and analyze how ethical considerations can impact the quality of a software product during the requirement elicitation phase.	**2**	(CO1, L4)
4)	Analyze the concept of a view in a relational database and differentiate between a base table and a view.	**2**	(CO3, L4)
5)	Analyze and differentiate between the physical and logical layers of database management systems.	**2**	(CO2, L4)

Section B **5*4=20**

6)	Construct a relational database schema for a Library Management System using DDL queries. Include at least three entity tables with appropriate attributes and describe how to create a view that shows the total number of books borrowed by each member. Provide SQL code snippets to define the entity tables and the view.	**5**	(CO3, L3)
7)	Describe and illustrate the process of requirement elicitation using an example from a healthcare application.	**5**	(CO2, L2)
8) a)	Explain and demonstrate how the conceptual, logical, and physical layers of a relational database management system (RDBMS) interact with each other using an example schema for a University Course Registration System.	**5**	(CO2, L3)

OR

b)	Examine and discuss potential issues that might arise during the requirement elicitation phase for a new e-commerce platform, and suggest strategies to mitigate these issues.	**5**	(CO2, L3)
9)	Analyze and differentiate between the process of requirement elicitation and the concept of quality assurance in software development, providing an example scenario for each.	**5**	(CO2, L4)

Section C **10*2=20**

10)	Construct an ethical requirement specification for a new mobile banking application that ensures user privacy and data security. Your response should address the following parts: (i) Identify potential ethical issues that may arise during the development of this application. (ii) Describe how you would conduct effective requirement elicitation to capture users' needs while ensuring their privacy is protected. (iii) Explain how quality assurance can be used in testing this application to ensure it adheres to ethical standards and user expectations.	**10**	(CO1, L3)
11) a)	Discuss the importance of understanding and designing various layers involved in software product development, focusing on their roles and interdependencies. Provide an explanation for each layer and illustrate how they collectively contribute to the overall quality and functionality of the software.	**10**	(CO1, L2)
	OR		
b)	Explain the concept of a relational model in database management systems. Discuss its significance in designing ethical software products. (i) Describe the key components that constitute a relational model. (ii) Analyze how the principles of a relational model can be applied to ensure data integrity and privacy in software development. (iii) Illustrate with an example how a specific aspect of the relational model, such as normalization or referential integrity, can address ethical concerns in real-world applications.	**10**	(CO1, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use